Tatemis3D · Elemental Reaction System

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In short: the terrain is **a third player with a history**. A **pure elemental hit only does damage**; *effects* are born only when it mixes with the terrain, in **five outcomes**. An impact **doesn't trigger anything on its own: it deposits saturation**. The **patch** charges, ends up **primed**, and the next hit pops it with a **wave that travels through it**, **transforms the ground**, and that ground **comes back on its own** after a while. Few events, big ones, and a board that changes with what the monsters do.

The base layer

One verb per element

Each element has **a signature personality**: not a list of effects, but **a verb** from which everything it causes is derived. The six sit on **different axes** and none overlaps with another.

ELEMENT	VERB	WHAT IT DOES	WHERE IT COMES FROM
Fire	**Spread**	Burns continuously, **spreads little by little** and ends up dying out on its own	fire burns *and spreads*; it has its own life cycle
Water	**Current**	**Drags** you while you are inside; when you leave, inertia keeps carrying you	water is a force that is hard to stop, not a shove
Nature	**Entangle**	**Blocks movement but lets you attack and block**	roots that grow tangled
Mineral	**Weight**	**Slows** (horizontal) and **anchors to the ground** (vertical: cuts juggles)	rock and crystal are the dense ones
Lightning	**Paralyze**	**Clock-driven ticks** that interrupt: they don't stop you moving, they stop you **chaining**	intermittent electrocution
Air	**Cloud over**	Turns a ground element into a **cloud**: it covers more area and **blocks vision**	air carries and suspends whatever it touches

> **The dominant element sets the VERB, the dominated one sets the MATTER.** A reaction carries the verb **of the element in charge**, and from the loser it only inherits **what it is made of**. That is why the sandstorm (Air › Mineral) blocks vision (Air's verb) and scratches because it is pulverized rock (Mineral's matter), but **it does not slow**: slowing is Mineral's verb, and there Mineral is not in charge. This rule is what stops the six verbs from contaminating each other.

The Nature ↔ Lightning mirror is deliberate: one takes away your movement and lets you hit; the other takes away your chaining and lets you move. They are in different triangles, so **they never dominate each other**; they only combine. **The Mineral ↔ Air mirror** too: one anchors and weighs down, the other suspends and disperses; and those two **are** a dominance pair (Air › Mineral).

> **Lightning's verb is the most systemic one:** the game's loop is *combo → continuous damage → stun → capture*, and stun decays if you stop hitting. An effect whose job is **to stop you chaining** doesn't take your health: it takes your capture.
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> **Ticks run on a clock, not on chance.** The reference is Pokémon's paralysis, but in real time a random interruption doesn't read as punishment, it reads as *a lost input*. With timed ticks the player **learns the rhythm and slips the combo in between them**: same feeling, and it rewards skill.

A pure element only does damage

Verbs are NOT properties of the loose element. Fire alone doesn't spread; Lightning alone doesn't paralyze. **Environmental effects are born exclusively from the mix with the terrain**, as in the real world. An elemental hit that doesn't touch elemental terrain is *damage of its type and nothing more*, and that is a valid, deliberate option: it is **safe and immediate**, and in a dominance duel the ×1.5 against the opponent is already its reward.

From there come **the five outcomes**, all different and all explainable in one sentence. This table says **WHAT** comes out of each pair; **WHEN** it comes out is decided by saturation.

WHERE THE HIT LANDS	WHAT HAPPENS	DAMAGE	EFFECT
Neutral ground	nothing, just the hit	your type	none
Your own element → amplification	your explosion **repeats** after a short delay (19 fire → another 19 fire)	your type, **twice**	none *(there is no mix to produce one)*
An element you beat → reaction	the reaction bursts, and it is **yours** (R6: it doesn't touch you)	**your element**	**your verb**
An element that beats you → terrain reaction	the same reaction bursts, but it is **its own** and **belongs to no side**	**the terrain's element**	**its verb**
Across triangles → combination	the area transforms	**neutral** *(of no element)*	**both verbs mixed**

> **Reaction = explosion · combination = transformation.** Choosing between hitting directly and aiming at the ground is a real decision: the direct hit is safe and immediate; aiming at the ground **invests** in a patch the opponent **sees filling up**, and when it pops it hits much harder but **takes time to reach them because it travels**. It works both ways: if you prime a patch under your own feet, that warning is yours too.
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> *Two verbs create an area by their very nature, **spread** (Fire) and **cloud over** (Air), so their reactions last instead of just bursting. It is deliberate: **the verb rules**, not symmetry.*

Elements also collide with each other

Mixes **don't need terrain**: two elemental deliveries that meet in the air resolve their pair **at the point of collision**. Ottorock throws its rock, Mantagale answers with a tornado, the two collide and the reaction is born there. It works for **any elemental delivery**, not just projectiles: a melee burst against an incoming projectile counts too. The rule is *two elements that touch react*, without listing cases.

- **It is impartial**: the mix comes from both of them, so **it affects both** (the same principle as the terrain reaction: nobody owns what they caused between them).
- **Same element → amplification**: the projectiles cancel out, but the explosion sounds twice.
- **The ground underneath doesn't count**: it already resolved once. Reactions **never chain** (R1).
- It gives elemental attacks a **defensive use**: a projectile can be *answered*, not just dodged. It is the interaction between ranged creatures.
- *Consequence for melee:* hitting up close leaves you inside the radius of an impartial mix. What protects you is hitting and getting out, and melee bursts being born **in front of the body**. Taking more environment is the fair toll of close combat, and part of why range exists.

> **Weight target:** a reaction is worth **almost as much as an average ability**. In a full duel, the split of total damage is **30% basic attacks / 40% abilities / 30% reactions and combinations**. The terrain **is not an extra**: it weighs as much as your basic attacks, and *a good combo consists of chaining basic attacks and abilities while aiming at elemental zones*.

The effect sticks

Effects happen in the area, **but they stay stuck to whoever they reach**. Inside the area the duration is **endless** (until the area dies); when you leave, the effect **decreases** until it fades. A burn keeps burning outside the blaze, less and less.

One charge instead of six rules: each effect carries a **charge that rises inside and falls outside**, and its intensity *is* that charge. The three properties come out of a single number: inside it holds, outside it decays, and going back in recharges it. And it throws in a fourth: **the longer it was inside, the longer it lasts outside**, so forcing someone to stay is the premium play.

This is what means **areas don't need CC to pay off**: pushing the enemy in for an instant already pays.

> **Two different things, two names. "Charge"** = the 0..1 intensity **stuck to the body** from this section. **"Saturation"** = what builds up **in the terrain**. They are separate systems and only share the word "build up".

Grace period

A live elemental effect **cannot be converted during a short grace period**. It covers two cases: **two effects from the same hit stepping on each other** and **the immediate bounce** inside a single detonation. Anti-spam is handled by the saturation thresholds: you can't spit effects onto a spot if each one costs 3 or 5 units.

> **Repeating the SAME reaction refreshes it; it doesn't get blocked.** The grace exists to prevent *switching* effects, not to punish persistence. Without this, Ottorock's signature mechanic breaks: its 3-hit chain in Grapple reproduces the sealed combination on every hit, and hits 2 and 3 would land on its own area in grace and do nothing.

Arena composition

Target cell proportion: **5 elemental / 5 neutral**. Each **biome** biases that mix toward certain elements: in a lagoon most cells are Water. **That is what rewards a versatile team**: the player chooses between specializing in a group of elements, covering a whole triangle, or mixing it up.

> **The environmental layer is a MODIFIER, not a cell class.** The second environmental system (ash, mist, snow) **is painted on top of any cell**, elemental or neutral. The board reads with **two** categories *(something happens here, or it doesn't)* and both environmental systems can cover 100% of the arena without taking space from each other.
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> **Neutral cells are not dead terrain:** they **charge at 2 units**, so they are **half the board in potential**. The arena that leaves you without the system doesn't exist: **you build it yourself**; it just costs twice as much as playing natural terrain.

> Causing reactions with an element you are weak to **is not always bad**: it is still damage to the enemy. Only the gameplay changes: your attacks start carrying risk for you.

Universal anatomy of an attack

Every attack in the game (basic attacks, abilities) has **three stretches**, and from there comes what can be stopped:

1. **Startup: interruptible.** Lightning ticks, hits and anything else cut it. **This stretch *is* the "I cut you mid-animation".**
2. **The hit: committed.** The movement of the hit + the hitbox. From here on **it lands no matter what**.
3. **Recovery: cancellable.** The player can **cut the tail with another ability** to chain more tightly.

> Three consequences: it gives Lightning **real counterplay** (it punishes whoever *starts* attacks, not whoever is already committed → the answer is to come in already attacking and leave no gaps) · cancellable recovery **is a combo system that feeds capture**, because chaining more tightly raises the combo multiplier and with it the stun · and Ottorock's charge has exactly this shape (wind-up → charge → slide).

Who each effect belongs to

A single sentence generates all five cases: the effect belongs to the element IN CHARGE; if nobody is in charge, it belongs to nobody.

SITUATION	WHO IS IN CHARGE?	DOES IT AFFECT YOU?
Reaction on an element you beat	**you**	**No**: R6 spares you

SITUATION	WHO IS IN CHARGE?	DOES IT AFFECT YOU?
Amplification (your own element)	**you**	**No**: it is your explosion again
Terrain reaction (it beats you)	the terrain	**Yes**, belongs to no side
Combination (across triangles)	**nobody**	**Yes**
Element collision	**nobody**	**Yes**

The difference between the two reactions is not *where* they happen (both are on the terrain) but *who wins*, and the triangle decides that. Ottorock (Mineral) hits a Lightning patch: Mineral › Lightning, its own element is in charge, the reaction is Mineral's and spares it. The same Ottorock hits an Air patch: Air › Mineral, the ground is in charge, Air's reaction comes out and doesn't spare it. **If your element is in charge, you control it; if the ground's is, you have woken it up.**

> **R6 protects you from the BURST, not from the AREA.** Your reaction doesn't hit you when it is born, but if it leaves an area that lasts (blaze, sand cloud, gravity), **that area doesn't tell anyone apart** and affects you if you walk in. That is why reactions with an area are more dangerous to use than instant ones, even when they are yours.
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> **Why combinations belong to no side:** if they spared you, placing them would *always* be good and there would be no decision. It is the risk to yourself that turns **where** you place it into a play. And it follows from the base principle: the terrain is **a third player**, and a third player doesn't take sides. A bonus: you can **ride your own current** or fight inside your own mini explosions.

The three sources of elemental effect

Outside these three **there are no effects**, only typed damage:

1. **The terrain:** 5 out of every 10 cells.
2. **The collision between elements**: in the air, with no ground involved.
3. **What a creature carries on itself: Ottorock's sealed rock**, the single exception to R5.

> Balance consequence: on neutral ground the game stays at damage + kit, **with no verbs**. That turns the **arena's composition into a first-order balance lever**, and it makes the crystal node and the sealed rock worth far more than they look: they are the two ways to have effects where the ground doesn't provide them.

Effect ceiling and dodgeability

At most 2 effects stuck at once, and the ceiling **is not the target**: the game aims for **one or none**. If the player positions well, they come out clean. **Everything has to be dodgeable**, and the 30% of damage the terrain contributes comes from **landing** reactions on the opponent, not from unavoidable environmental attrition. If being under an effect is the normal state, effects stop reading as events.

> **Rule:** *if an effect can't be seen on the body of whoever it affects, it doesn't go into the game.* It is what separates "the mud is punishing me" from "this game's dodge sometimes fails".

Terrain saturation: when everything happens

> **The terrain builds up; it doesn't react.** An elemental impact **doesn't trigger anything**: it deposits **saturation** in the patch it hits. When the patch fills up it is **primed**, and the next impact pops it (much stronger and much farther), **transforms the ground**, and that ground **comes back on its own** after a while.

Why. With half the ground elemental, one reaction per impact makes the default state *"something is always happening"*: blurred stimuli that **stop reading as events**. The answer is not removing terrain: it is **concentrating the same damage into fewer, bigger events**.

The threshold ladder

`u` **= one impact unit** = the impact strength of an average ability burst. A basic attack contributes ≈ **0.3 to 0.5u**.

WHERE IT LANDS	THRESHOLD	WHAT COMES OUT	WHAT IT LEAVES
Your own element	**1u**	**amplification**: instant, **no priming or delay**; it is the fast lane	nothing, but its radius **spills out of the cell** and deposits saturation in neighbouring patches
Neutral ground	**2u**	the cell **turns elemental**	a **single-cell** patch
Across triangles	**3u**	**combination**: boosted range and effect	**neutral ground**
Inside your triangle	**5u**	**reaction**: far more damage and far more range	**the by-product element**

The ladder **sorts by frequency and by power in the same direction**: what comes out most often does the least. Combinations are *transformations that last* and are the **texture**; reactions are *explosions* and are the **event**. The **1/2/3/5** ratio is fixed; the target is **one terrain event every ~15 s**, each one ≈ **4 to 5 times an average ability**.

It is measured in impact strength, not in damage dealt. Measured in damage, the charging speed would depend on the species' damage stat and the terrain would be a stat check. Measured in impact strength, which is authored per ability, **charging fast is a design decision**, not a side effect.

The battery is the PATCH

What builds up is the **whole patch** (the 3 to 6 connected cells of the same element), not the single cell.

- **Hitting it is easy**, so there is no need to make cells bigger.
- **The explosion IS the patch**, so "much more range" comes for free and **shaped like an amoeba**, not a circle: the board's geography goes from aesthetic to tactical.
- **The combat zone's fence CUTS patches.** Two neighbouring cells of the same element **are not the same patch if the wall runs between them**: *if you can't walk from one cell to the other without leaving the arena, they are not the same patch.* Design consequence: **where the fence falls changes the board, it doesn't just bound it**. A big patch split in two is **two small bombs instead of one huge one**, so placing the zone is a balance decision.

There is one meter, and it is contested

One meter per patch, owned by the element that is charging it. A different element **lowers it** instead of raising it; when it reaches 0 it **changes owner**, and the threshold with it. A single bar: readable, and it turns the terrain into a **contested objective**, which is the "third player" with a face.

Defending is much cheaper than stealing. Someone at 4u out of 5 is missing 1u; for the opponent it costs 4u to drain **plus** 3u to fill. Denying terrain is the last resort, not the reflex.

Grace, then decay. After the last hit the meter **holds** for a grace period (it doesn't move unless someone raises or lowers it) and **only afterwards** starts to drop. The grace gives time to reposition, chain and choose when to keep going; the decay keeps the whole board from ending up primed.

Primed, not automatic

When full, the patch **doesn't explode: it stays primed**, and detonates with **the next elemental impact**. The player chooses the moment and the terrain becomes a **combo finisher**, which is exactly what the base layer asks for (*"a good combo consists of chaining basic attacks and abilities while aiming at elemental zones"*). **Guardrail:** if nobody detonates it within **~5 s**, **it vents**: a weak flash, no effect, meter back to zero.

The detonation CHAINS

The patch **doesn't pop all at once**: **the cell that took the impact** pops and the **wave advances by adjacency** through the rest of the patch, leaving transformed terrain behind. It is the mechanism of the blaze reaction, generalized to all six elements.

- **The player AIMS the wave.** It is born where the hit lands: detonating from the near edge sends it away from you; from the far edge it brings it back on top of you, but it catches whoever is running. A new decision **with no new input**.
- **The wave IS the dodge window.** The time to get out of the way comes from **the travel**, and it can be seen.

- **It hits each creature ONLY once.** Otherwise, standing in the middle of a 6-cell patch would mean 6 impacts and the reaction would stop being an event and become a bomb. **It is seen travelling, it charges once.**
- **The wave moves faster than running**, or it gets outrun forward. It is dodged **sideways or backwards**: with a decision, not with speed.
- **REACTIONS chain; combinations SPREAD.** A reaction is an explosion: it travels. A combination is a transformation: **it occupies the patch**. Amplification chains nothing.

What it leaves behind

Every detonation changes the board: the patch takes on the by-product and the RING of cells around it gets stained with it. When the by-product and the patch match, **the patch expands**: there is no null case.

- **Reactions** → the by-product listed with the reactions. **Combinations** → **neutral** (nobody won; the ground got used up).
- **A detonation does NOT deposit saturation.** It is the payout, not an input. Without this rule, the ring it just created would get charged by the burst itself and there would be an **infinite domino**.
- **A detonation erases live effects within its reach**: the ground that held them stopped being that element.

Reversion with refresh

The transformed cell **returns to its baked state** after a cooldown, but **if there is fighting on top of it during that cooldown, the cooldown refreshes**. **Ground that is being fought over stays changed; ground that is abandoned heals.** Combat leaves a scar where it really happened, and the arena **never runs out**.

- **CHARGING refreshes it, not just detonating.** It is enough to **move the meter** (raising yours or lowering the opponent's) to reset the clock, so a patch you are charging doesn't revert halfway through.
- **Double the time on cells that were born NEUTRAL.** Building your own board is a **two-step** process (converting the neutral cell at 2u and *then* charging it to 3u or 5u), and with the same clock it would come undone halfway. Natural terrain is still better (cheaper, bigger patch), but manufactured terrain **gives you time to use it**.

A hit that doesn't fill it speaks, but doesn't act

Every impact that doesn't complete the patch produces **a silent flash of the right mix** (the steam hisses, the grass crackles) **with no damage, no status and no area**, and it **grows with the saturation**: almost imperceptible at first, loud when little is left. It does two jobs:

1. **It teaches the 15 pairs.** With mixes now rare, it is how the player learns what each combination produces.
2. **It is the telegraph.** A patch about to pop **can be seen and heard from afar**, and that is why the detonation can afford to be huge without being unfair.

Visible saturation is not polish; it is half the mechanic. If the hexagon can't be seen filling up, the player doesn't perceive charge: they perceive that hitting the ground stopped doing anything. That is why there are **two signals for two distances**:

- **Up close, the hexagon fills:** it grows from the centre and lights up with the colour of the element charging it. **Growing** is what reads as a meter.
- **From afar, the column:** a vertical bar in the centre of the patch whose **height is the meter**. It lets you decide *"I'm going to that zone"* without being on top of it.
- **When primed, both pulse white.** That pulse is what makes a huge explosion fair.

The three instant exceptions

They don't go through saturation and resolve on the spot (weaker than a detonation, in exchange for being immediate):

1. **Mid-air collisions**: there is no ground to build up in.
2. **Ottorock's sealed rock**: it is its signature mechanic and the only portable source of effect.
3. **Pyrecko's lit mines**: the minefield is still a minefield.

The three layers (don't confuse them)

① **Terrain: the board, and it has a history.** When combat starts, **an area is bounded**, divided into **hexagonal cells**. Each cell has **either an element, or nothing** (the environmental layer is a modifier on top), baked from the biome. **The terrain starts in a state and that state holds while nobody works on it**; a detonation transforms it, and reversion with refresh brings it back. The 6 elemental terrains:

ELEMENT	TERRAIN
Fire	volcanic rock
Water	river
Nature	grass / tall grass
Mineral	ore field (minerals embedded in the rock)
Lightning	electrified stone (like the volcanic one, but with electricity)
Air	air currents (geysers that blow out air instead of water)

All 6 elements have their own terrain on purpose: without Air terrain, the Lightning › Air dominance would be unreachable. **Neutral** cells don't react, but **they charge at 2u and turn elemental**. The world is continuous: the grid is an **overlay** that marks each section's predominant element, with a **framed combat zone**.

> *Separately, and unrelated to the type chart:* **physical and weather phenomena** (dust when hitting the ground, snow, rain, wind) are the game's **second environmental system**: physics that abilities exploit, without going through elements. It can snow without an Ice type existing. The two layers live in the same arena without touching.

② **Reactions: temporary, on top of the terrain.** What an elemental impact produces on terrain of another element. **They last a short time and go away.**

③ **Species marks: independent of everything above.** Things a specific species sows that **are neither terrain nor reaction**: Pyrecko's **"fuel"**, Mantagale's tornadoes. Their own rules are under per-species exceptions.

Resolution rules

R1 · Always against the terrain. An impact resolves against the **terrain's element right now**, never against an effect still active on top of it. *(Ottorock turns a stretch of river into **quicksand**; two seconds later Pyrecko fires there: the fire meets the **river**, not the quicksand → the terrain's **geyser**, which launches Pyrecko too.)* This prevents the combinatorial explosion: the matrix is 6×6, full stop. If a detonation turned the grass into ash, the next impact resolves against **Mineral**: **what changes is the cell, not the chart**. **It also applies to mid-air collisions**: they resolve once and the ground underneath doesn't count again.

R2 · Duration is proportional to impact. The harder the hit that radiates the element, the longer the reaction lasts. It rewards putting the heavy ability in the right place instead of spitting cheap abilities.

R3 · Attacking a live area DRAINS it. Hitting **inside** a live area with **another element drains its duration** (the same tug-of-war as the saturation meter). **That impact doesn't deposit saturation**: you spend the hit clearing instead of charging, so the counterplay keeps its cost. It has to land **inside**, not graze it: the point is aim, not proximity. On top of that, **a detonation erases live effects within its reach** without paying anything.

R4 · Directional rule. The same pair resolves differently depending on who comes in. The **predator** comes in → standard **dominance** (R6 protects its side). The **prey** comes in → **terrain response**: the **same reaction** from the predator's row fires, at full strength but **belonging to no side** (exception to R6). You can't burn what is wet: **what is wet vaporizes you**; putting out what is burning is still Water's normal dominance. *(In same-element mirrors, the terrain becomes the third player: both can provoke it and both must respect it.)*

R5 · Neutral surface → nothing. With no terrain element there is no reaction at the moment of the hit. **The single exception: Ottorock's sealed rock**, which carries its reaction already resolved and therefore also works on neutral

ground.

R6 · The burst doesn't hit whoever causes it (nor their side). The resulting **area effect** is neutral, though: it affects whoever walks in, from any side. *(Without this, Ottorock electrocutes itself at its own feet when grappling on Lightning, and a wild Ottorock would hand itself the knockout.)* **Two cases with NO side, not even in the burst:** the **terrain response** and **all combinations and collisions**. The full picture is under "Who each effect belongs to".

The 15 pairs: 6 reactions + 9 combinations

All 15 are derived from the verbs above, and each one has **a function none of the others has**.

The two triangles: ① **Nature › Water › Fire › Nature** · ② **Mineral › Lightning › Air › Mineral**. Inside a triangle there is always a **reaction**: it is *yours* if the predator comes in and *the terrain's, with no side*, if the prey comes in (R4). Between triangles, always a **combination**. And on your **own** element, **amplification**.

Each element has **2 triangle relationships** (who it beats, who beats it) **+ a signature** that explains its whole row:

ELEMENT	SIGNATURE	ELEMENT	SIGNATURE
Fire	burns; air stokes it	**Nature**	absorbs and spreads
Water	conducts, puts out, muddies	**Mineral**	hardens, crystallizes
Lightning	electrifies what it touches	**Air**	stirs, stokes and erodes

The 6 reactions: derived from the dominant element's verb

Each one deals **damage of the element in charge** (≈ an average ability) **+ its verb**. They fire both ways (R4): if the predator comes in it is *theirs* and doesn't touch them (R6); if the prey comes in, it is **the terrain's** and belongs to no side.

PAIR	REACTION	WHAT IT DOES = THE DOMINANT ELEMENT'S VERB
Nature › Water	**Explosive growth**: the roots drink the puddle	**Traps**: whoever gets caught can't move, but **can attack and block**. **Ground only**: it doesn't reach whoever is flying, which is the natural way out. It is the system's best **setup** tool: the only thing that pins someone inside someone else's area.
Water › Fire	**Geyser**: the water puts out the fire	**Launches**: hot water and steam that lift you up. It is the reaction that **opens up aerial play**, and what lets a Water or Fire melee reach a flyer. When you fall, the current's inertia keeps dragging you.
Fire › Nature	**Blaze**	**Spreads**: it catches and **extends little by little, only through Nature terrain** (the patch defines how far it goes) until it dies out on its own. Whoever touches it carries the burn with them.
Mineral › Lightning	**Geo energy**: the rock acts as a ground connection	**Weighs down**: an area of **stronger gravity** that pulls downward. It doesn't stop you taking off: it makes taking off **worth less**, shortening the hang time. It slows and pulls out of the air whatever flies → it is the game's **anti-air** reaction.
Lightning › Air	**Storm discharge**	**Cuts the combo**: clock-driven ticks that only bite during an attack's **startup**. It doesn't take your health: it takes your capture, because without a chain the stun decays.
Air › Mineral	**Sandstorm**: the wind erodes the rock	**Clouds over**: it lifts the rock into a cloud that covers far more area than it hit and **blocks vision**; it scratches because it is made of pulverized rock. **It doesn't slow** (that is Mineral's verb, and here Mineral is not in charge). It is the one **Ottorock eats**: on air currents, its own attacks blind it.

What terrain each reaction leaves

When it detonates, the patch takes on the **by-product** and the **ring** around it gets stained with it. Each result comes from the **physical phenomenon**:

REACTION	LEAVES TERRAIN OF	WHY
Nature › Water: explosive growth	**Nature**	the roots drink the puddle and stay
Water › Fire: geyser	**Air**	the steam
Fire › Nature: blaze	**Mineral**	the ash

REACTION	LEAVES TERRAIN OF	WHY
Mineral › Lightning: geo energy	**Fire**	the rock overheats with the charge
Lightning › Air: storm discharge	**Water**	the rain after the storm
Air › Mineral: sandstorm	**Lightning**	the suspended sand **builds up static** *(dust storms generate electric fields through friction)*

> **Parity is a balance rule.** The six elements appear **exactly once** as a result, so the table is a **permutation**: **no element can go extinct on the board or take it over**. With parity, **the board changes without drifting**.
>
> **All 9 combinations leave NEUTRAL.** Nobody is in charge, so nobody claims the ground: it got used up.
>
> **No case is null, thanks to the ring.** *Explosive growth* leaves Nature, and one of its two directions (Water comes in on Nature terrain) happens on ground **that is already Nature**: the patch doesn't change, but **the ring does**, so its function there is **to expand the patch**. The only reaction called *growth* is the only one that makes terrain grow. And it has a lovely read: **a Water creature fighting on grass waters its opponent's meadow**, exactly the pair where it loses.

> **Reverse trigger:** these 6 rows also fire **in reverse**: the prey comes in on its predator's terrain and the terrain answers with the **same reaction, belonging to no side** (R4/R6). *Fire on a river → **geyser** with Water damage, which also launches whoever fired; Mineral on air currents → **sandstorm**, which also blinds Ottorock.*

The 9 combinations: derived from BOTH verbs

Cross-triangle: **neither dominates**, so the damage is **neutral** and **they belong to no side** (they affect you too). They are **transformations**: areas that last, not bursts. Each one has **a function different from the other eight and from the 6 reactions**.

PAIR	NAME	UNIQUE FUNCTION	WHAT IT DOES
Mineral + Water	**Quicksand**	**Anti-escape**	Weight on a current = a dense fluid that **swallows you**. Inside, the dash and the jump come out **whole and at normal distance** (cutting the distance would read as a bug, not as mud), but **when you land you sink and the recovery gets longer**. Escaping works; escaping leaves you exposed.
Mineral + Fire	**Lava flow**	**The only area that ADVANCES**	Fire with weight on top doesn't jump from bush to bush: **it pours**. It is born at the impact and creeps slowly in the direction of the hit, eating up playable ground, until it solidifies and dies. Whoever touches it leaves with molten rock that keeps burning while it cools. **When it reaches a Water cell it goes out**: that is its death, and it is the opponent's answer. It is the only thing in the system that **gives distance back**. No side + advances = **it chases you too**: the most demanding one to use.
Mineral + Nature	**Crystallization**	**Takes away your CANCEL**	Entangle lets you act; Weight says that whatever starts doesn't stop. You fight, block and move **at normal speed**, but **you pay for every action in full**: you can't cut the recovery to chain. It doesn't take your damage, **it takes your capture** (without a tight chain the combo doesn't rise). The exact mirror of Lightning: *it cuts your hit, this one doesn't let you cut it.*
Lightning + Water ★	**Current** *(electrified water)*	**The only drag**	An electric conveyor belt: a **directional** sheet (the direction is set by your hit's vector flattened onto the ground, so **you aim it**) that carries whatever steps on it, with decreasing inertia when you leave. On top of that, Lightning's tick biting the **startup of your attacks**, never your movement. You can always get out, but you spend your escape leaving instead of counterattacking.
Lightning + Fire	**Mini explosions**	**Amplifies** *(the only one that raises other people's damage)*	One extra tick (Lightning) + damage spread over time (Fire): **every impact inside the area is followed by a mini explosion** that repeats a percentage of the original hit. Together with fog, **one of the two areas you WANT to step on**. No side, so whoever wins the exchange wins more. **The echo doesn't trigger reactions**: it already *is* the combination, and chaining would break R1.

PAIR	NAME	UNIQUE FUNCTION	WHAT IT DOES
Lightning + Nature	**Binding**	**Anti-kiting**	Lightning sets its **metronome** and Nature the **anchor**: every so many ticks, a **yank** pulls you back to the point where it caught you, each victim with its own stake. Between yanks you run normally, and your offence is untouched; it is *fighting to break free*. Discrete and not continuous **on purpose**: a constant force against your input reads as broken controls, a yank reads as an event. The longer you hold out, the weaker the next one, until the leash snaps.
Air + Water	**Fog**	**Misinforms**	A low, wide, opaque layer: **you can't LOCK ON through it**, and it hides whoever is inside just as well as whoever is looking from outside. **The other area you want to step on.** Projectiles already homing don't get lost: what is cut is the **acquisition**, not the homing.
Air + Fire	**Fuse**	**Timer** *(the only one that does nothing while it lasts)*	Clouded-over fire doesn't burn: **it builds up**. The cloud grows and saturates its whole life without dealing damage, and **deflagrates all at once in its entire volume at the end**, air included. That is why it is **the only one that reaches a flyer**, but as a one-off event you can walk away from. **It detonates only by its clock**, never by hits, or Pyrecko would get a free detonator on top of its mark kit. Here the dangerous part is the end.
Air + Nature	**Spore veil**	**Interdicts what is thrown**	The tangle leaves the ground and stays **hanging**. You can walk, jump or dash through it with no effect: what it eats is **the elemental projectile that crosses it**, which **detonates there**. The veil behaves like a stationary element for the collision rule. It is the **natural counterplay against a Zoner**. It only detonates **what is elemental**: firing without an element passes through it.

> **Ottorock's menu.** The three Mineral ones are the ones it seals in Grapple, so **the player chooses between them**. That is why they have **very different radius and duration**: crystallization small and long · lava flow narrow and directional · quicksand wide and short. That way the choice is real and its signature mechanic isn't decorative.
>
> **Stun ≠ stun-lock.** "Stun" fills the **capture bar**; it is not a control effect that takes away your input. The real CCs (trap, drag, yank) are called by their name.

Per-species exceptions

Ottorock: portable sealed combination. It is its signature mechanic in Grapple: when it **grabs** the terrain, Mineral + the terrain's element resolves **once** and the result stays **sealed in the chunk**. From then on **every impact reproduces that same mix**, on any surface, **including neutral**. It is the game's **only portable source of effect**. It chooses its combination by choosing **where it grabs**: river → **quicksand** · volcanic rock → **lava flow** · grass → **crystallization**.

- **Instant: it doesn't go through saturation.** Every impact reproduces its combination **on the spot**, without charging any patch. It is weaker than a detonation, and immediate in exchange.
- Neutral terrain when grabbing → **pure rock**: the chunk reacts as **Mineral** against the terrain where it impacts.
- **Only what is a COMBINATION gets sealed.** If the grab triggers a **reaction** (on Lightning, which it dominates; on Air, which dominates it) or an **amplification** (on its own ore), it is used up there and the chunk stays **pure**.
- When it goes back to Free (breaking or throwing the rock) **it loses the combination**.

Pyrecko: the "fuel". A species mark, **not** elemental:

- **Inert**: only its Signature **lights** it, and lighting it **doesn't detonate it: it arms it**. A lit mark **bursts into Fire at the slightest contact**: **Fire damage** to whoever touches it, and **the burst radiates Fire against its cell** → that fire resolves normally against the terrain (on Water terrain → **terrain response**: the **geyser**, with no side and **Water** damage, which launches Pyrecko if it is inside). The lit field is a **minefield**: *not stepping on it* is the counterplay.
- **Its kit applies no effect on its own**: everything it produces is **Fire damage + Fire delivery**; effects are born only from the mix.
- **It only disappears when it expires (10 s) or when it bursts**, and it bursts **only once**: once lit it is a **single-use mine, consumed on contact**, so the field gets used up as it is stepped on.

- **The lit mine is instant: it doesn't go through saturation.** When stepped on, the burst resolves right there against the terrain. If it deposited saturation it would stop being a threat and become a board-charging machine.
- **It belongs to the place, not the material:** if Ottorock tears up the chunk of terrain where the fuel is, it takes **the terrain, not the mark**.

Mantagale: tornadoes that push and detonate on collision.

- Its **tornadoes push along their whole path** (area denial), but they are **elementally inert**: they don't touch the terrain or trigger a reaction while travelling. Just like Pyrecko's fuel, they are a species mark with its own logic.
- The reaction happens **when they pop**, and they pop by **collision of opposing forces**: the tornado pushes, its Cyclone **pulls**, and the clash detonates them. The reaction resolves against the terrain **at the collision point**, with the enemy dragged in there.
- The Cyclone **drags the enemy along the ground** to the centre (it doesn't teleport it): on the way back, **it passes through the tornadoes sown along the path**.
- Its strength **depends on the arena's terrain**: it is the one that "plays the board". On **neutral** it achieves nothing. On **Lightning**, when its tornadoes pop they cause the **terrain's storm discharge, with no side**, which also cuts the **startup** of its own attacks: it is its **high-risk** board.
- **Spirit fish: the fuel for its flight.** Every detonation it causes gives birth to **5 fish at once**; they travel to it and **are consumed one by one, quickly**. **The fish is not height: it is lift.** Mantagale **always glides**, and the fish **flip the sign of the glide**: with fish it rises, without fish it descends. At the ceiling they aren't wasted: they buy **staying power**, so **landing everything keeps it up**. **One detonation = one flight**: its height is an **arc**, it shoots up and comes down setting up the next one.
- **The fish count as PROJECTILES:** an **elemental attack** destroys them **in transit**; neutral ones don't. Its height is **an attackable resource**, and the **spore veil** kills them, including its own. **The 5 travel along different curves**, so they can't be popped in one pass: killing 2 of them = it rises 3 steps. The counterplay is a **gradient**, and its window is **the journey** (consumed = untouchable).
- **The collision that pops a fish does NOT give it another one:** the mix born there **belongs to nobody**. Whoever fires pays an **impartial** flash for each fish.
- **Only the detonations IT causes give it fish**, not its teammates'.
- **Its gusts (basic attack) are NEUTRAL unless boosted:** they are only an **Air delivery** when the Flex boosts them. Otherwise the spore veil it creates on grass would detonate its own basic attacks on top of it.

Readability

- **Reaction = loud (explosion) · combination = soft (transformation) · terrain response = the VFX of its mirror reaction with a "provoked" nuance** (it starts from the terrain toward the attacker, not from the impact outward). The only "nothing happens" cases are **neutral ground** and, only as far as effect goes, **amplification**, which does hit.
- **A reaction has to separate in luminance from the terrain where it is born.** It is a hard constraint, critical for combinations, which have no flash.
- **If an effect can't be seen on the body of whoever it affects, it doesn't go into the game.**
- **Every reaction warns before it exists:** the warning is **the patch filling up in plain sight** and then **pulsing primed**. It lasts as long as the fight takes to charge it, and that is why the detonation can afford to be huge without being unfair.
- **The cell's saturation IS interface, not decoration.** The hexagon has to **be seen filling** and the primed patch has to **be seen and heard from afar**.
- **The silent flash grows with the charge**: it is at once **the permanent tutorial for the 15 pairs** and **the telegraph**.
- **The detonation wave reads as a front that travels**, not as a flash: that is what makes it dodgeable **sideways**, and what communicates that it was born where the hit landed.
- **Transformed terrain is distinguishable from baked terrain** (an edge or a slightly different tone that **fades when it reverts**), or the player wouldn't know that the ash on that patch was made by them.
- Effects **disperse on their own** (R2) and the terrain comes back: **the world changes, but it never stays painted.**